



When Using AI in Scientific Research: Start with Human, End with Human

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Abstract

Researchers are increasingly encouraged to adopt AI tools, yet there is still a lack of practical guidance on how to use AI without compromising the human essence of scholarly work. This paper addresses this gap by proposing a “start with human, end with human” approach to AI integration in research. At the core of this proposal is the Human–AI–Human cycle, a framework that maintains human agency at both ends of the research process. The cycle begins with the human scholar identifying problems, and formulating research questions and design, which all are grounded in human values and expertise. AI is then employed to assist with data processing, simulating, analyzing, and suggesting. Finally, the process returns to the human scholar, who handles the interpretation, sense-making, ethical review, peer dialogue, and public dissemination, all of which require human agency and accountability. This balanced approach ensures that AI amplifies rather than diminishes human scholar capabilities. It encourages scholars to embrace AI not as a threat, but as a tool that, when used wisely, can enhance, not erode, the soul of scientific inquiry. The implications are significant for research ethics, scientific practices, and the evolving definition of academic labor in the age of AI.

Keywords Artificial intelligence · Academic integrity · Writing · Research ethics · Human-AI collaboration · Scientific research

Prologue: Embrace AI, But How?

There is much discussion about embracing AI in academia, yet there is a noticeable gap: we often hear vague calls to use AI, but there are few concrete guidelines or examples on how to do so effectively. We are living in a unique time where AI is increasingly influencing the way we engage with scientific research. This presents an incredible opportunity, but also a challenge: *how can we use AI to enhance human potential without losing the deeper, human essence that makes research meaningful?*

Research can be broken down into two parts: *re-*, meaning *again*, and *search*. So, at its core, to research is to *search again*: to return, to look deeper, and to seek more thoroughly. Research has always been about the human pursuit of truth. It involves curious minds asking questions, exploring ideas, and building on evidence. With each new question, we

revisit old knowledge, challenge assumptions, and uncover insights that shape our understanding of the world.

With the rise of AI, particularly large language models like ChatGPT, the way many researchers draft, edit, and conceptualize their work has changed significantly. AI tools offer speed, clarity, and support, but they also come with risks. In this paper, I argue that when using AI in scientific research, we must *start with human, end with human*. By doing such, we ensure that AI serves as a tool to amplify the human scholar in us, rather than replace the critical thinking, creativity, and judgment that only humans can provide.

Start with Human

To *start with human* means to root writing in human experiences, curiosity, and intellectual rigor. While AI may excel at synthesizing vast amounts of information, it cannot mimic the questions born from a night of restless contemplation or the sudden insight that challenges an entrenched theory. These deeply human questions—“*What if?*” “*How did this happen?*” and “*Why?*”—often arise from personal experiences,

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societal struggles, or global concerns. AI cannot experience wonder or doubt, which often propels our most profound ideas.

Moreover, starting with human means understanding the context. A literature review assisted by AI may capture what's published, but it may miss what is urgent (Giray, 2023). Only researchers can recognize the gaps in knowledge and pose the critical questions that have yet to be asked. Only a human scholar can say, "This hasn't been asked yet, and it matters." After which, AI may help. Researchers may use AI to enhance the research process, but AI cannot and never replace the foundational human element behind research (Giray, 2023).

During the writing process, AI proves useful in refining language, improving structure, and helping overcome language barriers (Altmäe et al., 2023; Kacena et al., 2024), particularly for multilingual or non-native English researchers (Giglio & Costa, 2023). However, a growing concern in academia is the criticism directed at those who use AI, an emerging practice referred to as *AI shaming*. AI shaming is defined as "criticism aimed at individuals or organizations for using artificial intelligence in performing tasks" (Giray, 2024a, b, p. 6). "This phenomenon often involves dismissing the validity or authenticity of AI-assisted work" (Giray, 2024a, b, p. 2). Even when AI is used ethically and merely as a tool to enhance human capabilities, some assessors still lower their evaluations (Tan et al., 2025) or outright reject submissions (Faems, 2025) upon discovering that AI tools were involved in the writing process. Concerns about academic integrity and quality control are valid and should not be dismissed, yet indiscriminate AI shaming risks discouraging its thoughtful and transparent integration.

In fact, recently, anecdotal evidence has highlighted this tension: a researcher received a desk rejection from a top-tier journal after openly disclosing the use of generative AI for grammar support, literature prompts, and framing ideas (Faems, 2025). Ironically, the rejection was not due to the content or quality of the work, but solely because of the AI assistance in terms of idea generation. It was deemed a violation of the journal's policy.

If this had been a human work, simply accompanied by honest transparency about the responsible use of AI, it should have been seen as an ethical practice. Penalizing such openness sets a troubling precedent: it discourages researchers from being transparent about their methods and may incentivize concealment instead. This is not a healthy approach if we want to foster innovative academic practices.

Work with AI

AI demonstrates remarkable capabilities across multiple research domains. It excels in automatic file processing (Barr & Shaw, 2018), efficiently analyzes vast amounts

of data, and supports intelligent decision-making to enhance research performance (Himeur et al., 2023). In predictive analysis, AI uncovers real-time patterns, correlations, and opportunities, empowering researchers with timely and accurate insights (Zong & Guan, 2024). Given its deep grounding in both traditional statistics and contemporary data science, AI is increasingly being integrated into modern research workflows (Xing, 2024).

AI's usefulness also extends to qualitative research. Morgan (2023) explored ChatGPT's performance in qualitative data analysis and found that it performed reasonably well overall. Prescott et al. (2024) compared human and generative AI coders in thematic analysis, finding a promising consistency between the themes generated by both. This suggests that AI has potential to reduce the resource intensiveness of qualitative work.

Furthermore, advanced AI models have shown impressive alignment with reference standards in thematic analysis, with some even achieving perfect concordance and consistent performance across different subgroups (Bennis & Mouwafaq, 2025). In the areas of data recovery and analysis, AI shows great promise. Deep learning models are now able to retrieve image and video data by identifying patterns, reconstructing corrupted content, and assisting in the restoration of lost information (Singh et al., 2024).

However, these strengths are not without limitations. Prandner et al. (2025) highlighted challenges in using ChatGPT-3.5 for certain research tasks, noting that the model tended to repeat responses in loops when solutions were unclear. More importantly, Goyanes et al. (2025) argued that despite its power, ChatGPT lacked the capacity to replace the contextual insights, metaphorical sensitivity, and reflexive interpretation that human researchers bring. AI cannot grasp the layered meanings and subtleties that are often essential in rigorous analysis. Therefore, human researchers must remain central to the process.

The fundamental principle is this: work with AI, not through AI. Working *through* AI implies passive delegation. Working *with* AI, however, means engaging in an active partnership, where the researcher maintains authority and responsibility over every aspect of the research. Every suggestion, output, or insight produced by AI must be examined, validated, and ethically contextualized by a human researcher. Researchers cannot simply blame AI when something goes wrong or claim ignorance when errors arise. Accountability remains human.

Despite all the assistance AI can offer, it must never replace the human elements essential to meaningful inquiry. This is the balanced approach we must adopt.

End with Human

To *end with human* means reclaiming authorship in the final stages of research. Before submitting a manuscript, scholars should pause and reflect: *“Is this truly our voice? Our interpretation? Our unique contribution to the field?”* Ending with human serves as a reminder to critically evaluate the work that AI has helped refine. It is in this final reflection that we address the ethical implications of using AI in research: checking for accuracy, bias, originality, and alignment with our core research values (De Angelis et al., 2023). It is essential to maintain transparency about the role AI has played in the research process. Just as researchers disclose methodologies, they must also disclose the AI tools they utilize. They must acknowledge both the assistance and the responsibility that ultimately lies with them (Beutel et al., 2023).

The phenomenon of hallucinations in AI-generated content, where the text deviates from the intended meaning or misrepresents facts, further underscores the need for human oversight (Beutel et al., 2023). Such *hallucinations* can lead to flawed knowledge if researchers do not properly check the assistance provided by AI (Giray, 2023). These inaccuracies erode trust in research and contribute to the wider infodemic threat within scientific discourse (De Angelis et al., 2023).

There were also some unfortunate incidents where researchers failed to involve human oversight in the final stages of the research process, leading to problematic outcomes. For example, a team of materials engineering researchers used ChatGPT evidently in the introduction of their paper, which began with the phrase, “Certainly, here is a possible introduction for your topic” (Zhang et al., 2024). Similarly, in a study by medical researchers, the discussion section included an AI-generated response stating, “I’m very sorry, but I don’t have access to real-time information or patient-specific data, as I am an AI language model” (Bader et al., 2024). These papers were published but later retracted, thanks to the efforts of concerned scholars and the viral spread of the issue on social media, which led editors to retract the papers.

However, AI need not be seen as an adversary. As Eaton (2023) discussed, post-plagiarism ethics have emerged in response to the convergence of AI, neurotechnology, and brain-computer interfaces; these are the tools that reshape how we write, learn, and share knowledge. The question is no longer whether AI should be used, but rather how it can be used responsibly and ethically. Indeed, AI can foster a collaborative dynamic between human insight and AI capabilities (Davenport & Miller, 2022). As Dede et al. (2023) emphasized, AI augments human intelligence and offers vast opportunities when used thoughtfully.

AI can assist in the research process by facilitating cognitive offloading, reducing mental strain by delegating routine tasks (Armitage et al., 2020). However, writing is much more than just a collection of words (Arapoff, 1967). As Baron (2023) observed, writing is inherently a thinking process, which is fundamentally human. Therefore, AI should enhance, not overshadow, the reflective human aspect of research.

The Human-AI-Human Cycle

Some claim that AI can independently conduct research or even produce PhD-level dissertations in minutes, but this is simply not true. Early tests reveal major shortcomings: AI lacks contextual understanding, misses recent developments, fabricates information, and cannot reliably distinguish between credible and unreliable sources (Ciriello, 2025).

Many publishing houses and academic institutions still do not recognize AI as an author (Goto & Katanoda, 2023). It is not considered an author because it lacks agency and cannot take responsibility for the work (Thorpe, 2023). Hence, AI cannot produce research on its own. This stance highlights a key principle in research: the value of scholarly content lies not only in its structure, but in the creativity, critical thinking, and accountability that only human researchers can provide. Indeed, scientific integrity remains a distinctly human responsibility (Giray, 2024a, b).

At its core, the human–AI–human cycle (see Fig. 1) suggests that AI tools are inseparably linked to human intentions, ethical considerations, institutional norms, and collaborative practices. For example, research should (1) *start with the human*: problem identification, research question formulation, methodology design, and data selection must be grounded in human values and disciplinary expertise. (2) *AI can then play a role in the middle*: processing, simulating, analyzing, and suggesting, within the parameters and prompts provided by the human scholar. (3) *The process must finally end with human*: interpretation, sense-making, ethical review, peer dialogue, and public dissemination, all of which require human agency and accountability.

To ensure responsible and effective integration of AI in research: AI tools should be designed for complementarity, meaning amplifying human research functions. Researchers must maintain continuous alignment, regularly evaluating how AI influences their decisions, biases, and interpretations. And lastly, the human scholar must remain in the loop throughout the process, not only as a user, but as a steward of meaning, ethics, and purpose.

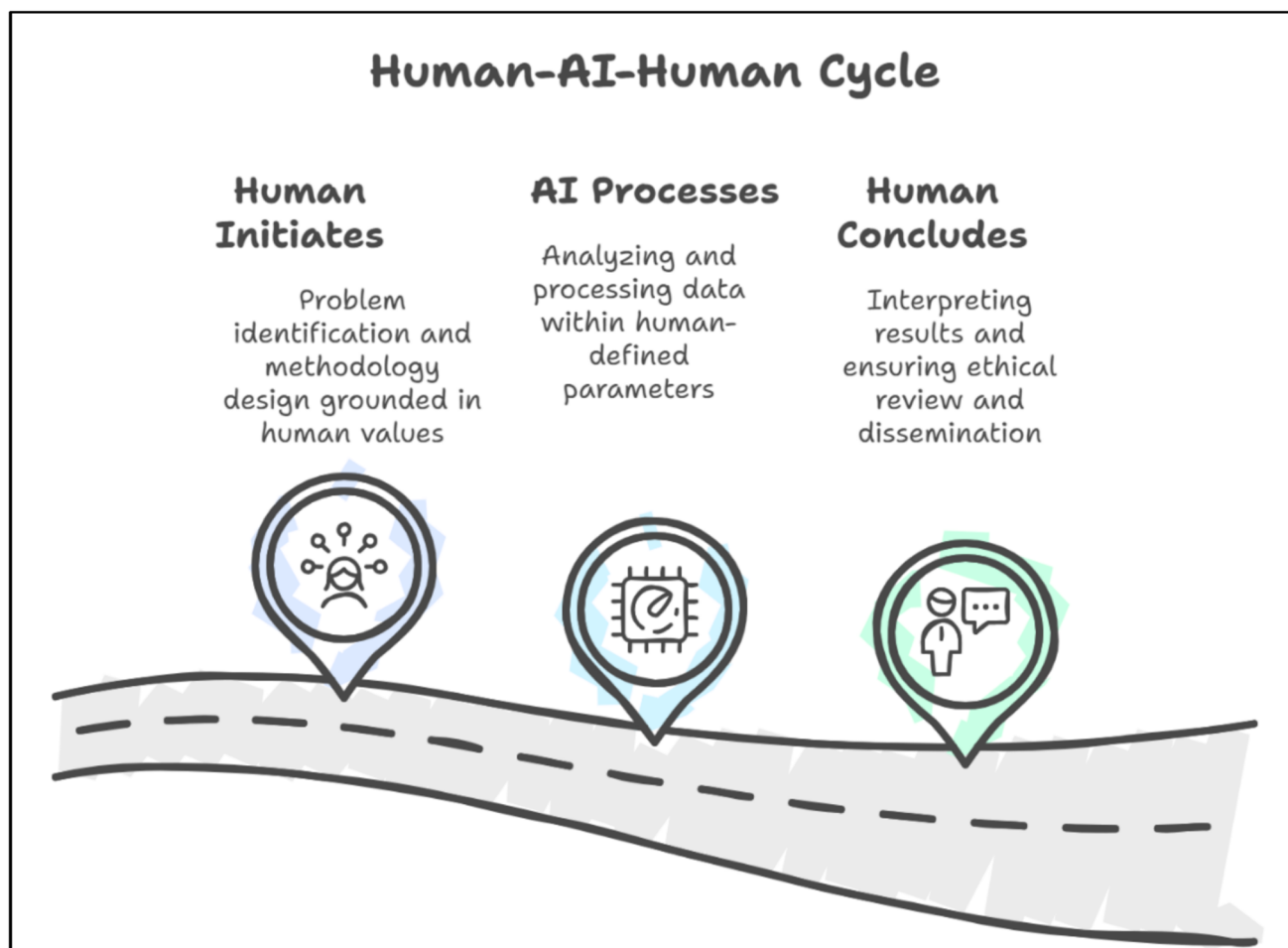


Fig. 1 Human-AI-Human Cycle

Implementation Strategies Across Research Contexts

For individual researchers, it is pivotal to establish clear guidelines regarding AI use: defining when, why, and how AI tools will be employed, as well as methods for verifying outputs. Maintaining a detailed research log to document AI contributions and limitations, alongside incorporating peer feedback, promotes transparency and accountability. Developing skills in prompt crafting, critical evaluation of AI-generated content, and thorough documentation further support responsible AI integration (Mollick, 2024).

In team and institutional settings, structured approaches are vital. Research teams should agree on shared standards, assign responsibilities for validating AI outputs, and implement peer review systems to ensure clarity and quality. Regular training and open communication help maintain effectiveness (Weller, 2011). Institutions must create formal policies embedded in ethical frameworks, provide training addressing both technical and ethical dimensions,

and establish infrastructure for monitoring AI use and auditing research integrity. Disciplinary communities also play a crucial role by developing field-specific guidelines, updating peer review processes, and fostering ongoing dialogue to ensure AI serves as a facilitative tool rather than a substitute.

Recognizing Adequate Versus Insufficient Human Oversight

Effective research in the age of AI hinges on sustained human oversight that ensures both ethical integrity and scholarly rigor. Adequate oversight begins in the planning phase, where researchers must be able to justify their study's purpose independently of AI-generated rationale. As Mittelstadt et al. (2016) emphasized, ethical AI use starts not with retroactive justification but with intentional, human-led design. The researcher's role involves deciding when AI is helpful and, just as crucially, when it is not. This

discernment must be backed by clear protocols for validating AI outputs.

Active monitoring is another core indicator. Oversight does not mean feeding data into an AI model and uncritically accepting the results. This means engaging critically with those outputs. Responsible researchers constantly assess whether AI remains aligned with research goals and step in when outputs become questionable. Katzenbach and Ulbricht (2019) framed this as critical engagement rather than passive consumption. Furthermore, strong oversight is reflected in the final output: insightful, contextualized, and theoretically grounded. AI alone cannot produce them.

In contrast, insufficient oversight is marked by overdependence. When AI dictates every stage from research question to interpretation. When this happens, quality suffers. Without human verification, errors and contradictions go unchecked. This leads to shallow, generic work lacking disciplinary nuance. As Noble (2018) warned, AI outputs are not neutral; they carry biases that must be interrogated. Theoretical contributions, in particular, often disappear under the weight of AI's remixing tendency.

Recent literature emphasizes the potential of AI-human collaboration when properly managed. Mollick (2024) underscored that while AI can automate routine tasks, the real value lies in leveraging AI to enhance human creativity and decision-making. Przeglasińska et al. (2025) echoed this, suggesting that AI is most beneficial when aligned with human strengths in a collaborative framework that supports innovation. In education, Järvelä et al. (2023) found that AI used as a supportive partner reinforces human cognition.

Assessment Guide for AI-Assisted Work

In the age of AI, research is no longer judged solely by its output but by the intentionality and integrity behind its creation. This assessment guide offers a reflective framework for assessing AI-assisted work, focusing on how researchers think, act, and remain accountable throughout the process.

Phase 1. Human Foundation

Before any prompt is written or algorithm engaged, the first question to ask is this: Has the human work been done? Every meaningful project begins with a solid foundation. A research problem must come from a real gap in the field or an issue that matters in the world. It should reflect intellectual curiosity, contextual awareness, and disciplinary depth. This cannot be supplied by AI. It requires the human researcher to identify, justify, and commit to the problem space.

At this stage, it is crucial that the research objectives, conceptual framework, and rationale are clearly authored by the researcher. They must not merely follow what a chatbot recommends. Has the researcher demonstrated deep engagement with the literature? Have they constructed their own point of view? Is the research direction shaped by the tool or by the scholar?

Ethical reflection also belongs here. As Brännmark and Sahlin (2010) argued, ethics is a precondition, not an afterthought. Assessors must look for early awareness of possible consequences, including bias, risk, or harm. Does the researcher acknowledge their own assumptions? Are limitations clearly stated? Has the ethical dimension been meaningfully considered?

Questions to ponder:

- Is the research problem grounded in relevance and disciplinary value?
- Are the research aims and frameworks crafted by the human researcher?
- Has the literature been personally read and understood, not just retrieved?
- Are ethical considerations present from the start?

Phase 2. AI Integration

Once the foundation is laid, AI can be thoughtfully introduced. The role of the assessors is not to look for the absence of AI, but to assess the appropriateness and intentionality of its use. AI should be used to amplify, not substitute, human thinking. As Weller (2011) stated, technology should extend human capability. It must not erase it.

The key here is complementarity. Is AI used to enhance human creativity, analysis, or clarity? Has the researcher selected the right tool for a specific task? Is the tool used to support discovery, synthesis, or design in ways that would otherwise be difficult or inefficient?

While notions of strict human-AI boundaries are fading, the responsibility remains. In research, AI assistance may be integrated so well into human reasoning that boundaries blur. That is acceptable, if the researcher remains in control. The important question becomes: Is the final logic still human-led? Is there reflection, refinement, and critical decision-making?

Transparency is essential. According to George and Wooden (2025), responsible innovation requires openness at every level. Has the researcher disclosed what AI tools were used and for what purpose? Have prompts, outputs, and iterations been documented? Has the researcher cross-checked or validated results, especially given AI's tendency to hallucinate?

Questions to ponder:

- Are the AI tools used to extend the researcher's thinking or skill?
- Does the use of AI match the task in purpose and precision?
- Has the researcher stayed in control of the decisions and logic?
- Is there honest transparency about which parts were AI-assisted and how?

Phase 3. Human Synthesis

This phase is where everything comes together. Human judgment must also take the lead here. A strong synthesis demonstrates that the researcher is not simply curating what AI produced. They are making meaning. They are interpreting, connecting, theorizing, and deciding what matters.

AI might offer drafts, lists, citations, or language improvements. But only a human mind can determine what is worth keeping, what is flawed, and what must be expanded. As O'Neil (2016) reminded us, technology is not neutral. The researcher must have applied critical eyes to filter inaccuracies, surface nuance, and reject bias.

The final synthesis must reflect original thought. Ideas should be framed in light of context and theory. Evidence should be woven with intention. Conclusions should emerge from logic, not convenience. At this point, assessors must ask: Is the researcher thinking? Or are they simply arranging?

Equally important is transparency. Has the researcher acknowledged the role of AI in the process? Have they explained what was generated, suggested, or influenced by a tool? Have they discussed findings with mentors or peers? As Boddington (2017) points out, ethical research is about trust. The reader must be able to trust not just the results but the process.

Questions to ponder:

- Does the synthesis show human meaning-making and not just arrangement?
- Are theoretical and contextual linkages clearly developed?
- Has the researcher filtered, questioned, and refined AI-generated suggestions?
- Is the researcher accountable about AI's role?

Moving Forward

Despite having a framework for responsible AI integration, we must look ahead to understand how human roles will transform and what new ethical considerations will emerge in this partnership. As AI increasingly automates routine

analytical tasks, human researchers are freed to engage in deeper cognitive functions (e.g., theorizing, pattern recognition across domains, and the creation of interdisciplinary linkages) that transcend formulaic scholarship. However, this promise is not without its perils. Many experts express concerns that AI might diminish individual autonomy in research (Giray, 2024b). This weakens scholars' control over their academic choices and direction.

The rise of AI unfolds within a broader neoliberal framework where technological advances are often portrayed as uncritically positive. Many dub it as a *productivity miracle* masking the structural rot within academia. As Fleming et al. (2021) noted, many technological innovations, while often framed as progress, may contribute little to meaningful insight within academia or broader society. The emphasis on speed and efficiency in research, particularly through AI, risks prioritizing output over reflection and deeper understanding.

This evolution demands that researchers not only adapt but also develop refined competencies in evaluating AI outputs critically, understanding algorithmic blind spots, and asserting intellectual sovereignty over their work. The researcher of tomorrow must shift from data processor to *knowledge orchestrator*. The research must adept in formulating research problems, generating original hypotheses, and synthesizing AI-assisted insights with human intuition and domain-specific wisdom. Yet, this transformation happens amid growing concerns about declining academic integrity, as scholars raise red flags on cheating, data fabrication, and an erosion of creativity and critical thinking (Giray et al., 2024). These issues speak to a deeper crisis of trust in academic labor, exacerbated by unregulated AI adoption.

Equally pressing are the ethical implications. As AI becomes more embedded in scholarly workflows, our long-standing definitions of academic integrity must evolve. The traditional paradigms, designed for human-only authorship, no longer capture the complexity of human-AI entanglements. New standards of transparency must be developed. This requires researchers to disclose the scope and nature of AI involvement, explain how decisions were made, and demonstrate rigor in quality control (Kocak, 2024). Without such revisions, we risk normalizing questionable practices and undermining scientific integrity.

Moreover, technical limitations persist. ChatGPT and its cousins may produce insightful outputs, they should not be solely relied upon for scholarly referencing due to documented inaccuracies (Giray, 2023). Similarly, widely used AI writing aids such as Grammarly may subtly erase stylistic nuances, raising important cautionary notes on overreliance and the homogenization of voice (Giray, 2024a). Thus, responsible AI use must be anchored in critical human judgment, not blind automation.

The institutional dimension cannot be ignored. Successful AI integration depends on coherent policies, robust training for educators and students, and a clear emphasis on writing as a process rather than merely a product. Without these, we risk turning to AI for policing misconduct rather than cultivating integrity (Giray et al., 2025). Compounding these issues is a disturbing trend: the emergence of *publishing addiction*, where generative AI accelerates writing but encourages a quantity-over-quality mindset that aligns with neoliberal university metrics (Giray, 2024b).

AI promises democratization, yet its uneven accessibility may deepen existing inequalities. Institutions in the Global South or with limited resources may be disadvantaged as AI becomes a prerequisite for cutting-edge research. This creates a digital divide, where the capacity for scientific contribution is concentrated in well-funded centers, marginalizing already underrepresented scholars. Moving forward, we must collaborate across sectors (e.g., researchers, institutions, technologists, and policymakers) to ensure AI augments human creativity, insight, and purpose in scholarly inquiry.

Epilogue: Soul of Science

This human-AI-human cycle is about upholding scientific standards, yet it is also about protecting the soul of science. Research is more than a process of producing knowledge. It is a way of making meaning. Over-reliance on AI risks flattening the emotional and intellectual richness that defines scholarly work. But when used mindfully, AI can stretch our capacities without compromising our core identity as human researchers.

If we want the Human-AI-Human cycle to work long-term, we need to keep showing up, not just as users, but as thinkers. It is about safeguarding human agency, staying vigilant with quality control, and leading innovation with integrity. AI might boost our potential, but it should still our voice, our choices, and our ethics that shape the scholarship. Again, for AI to be used wisely and effectively, remember to: *start with human, end with human.*

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Declarations

Declaration of Generative AI in Scientific Writing This manuscript used ChatGPT 4.0 solely for proofreading and editing purposes. All content in this publication is my own, and I take full responsibility for it.

Ethical Approval This paper is a reflective essay and did not involve human participants or empirical data collection; therefore, no ethical approval was required.

Conflict of interest The author declares no conflicts of interest, financial or non-financial, in relation to the authorship or publication of this essay.

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